Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Ctrl + Me

Selected Domain

Domain 2 - Orbyte Smart SME Challenge

Selected SDG Track (1–17)

SDG 8 - Decent Work and Economic Growth

Project / Solution Name

SupplyPilot AI

Name (Leader)

WANG MINMIN

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

1 (Solo)

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Across Malaysia, smaller bakeries, cafés and other micro-SMEs often manage purchase orders in spreadsheets and supplier follow-ups in chat messages. Before a delivery, staff must manually compare informal confirmations with the ordered products, quantities and delivery dates. One message may partially confirm flour, omit butter, or indicate that sugar is unavailable. These details are easy to miss when operations are busy.

A late or incomplete delivery can interrupt production, create urgent purchasing work and increase the risk of wasted ingredients. Owners also need to know which discrepancies were identified and which follow-up action was approved. Generic chatbots may summarize a message but do not necessarily reconcile it with a purchase order or provide a controlled next step.

SupplyPilot AI focuses on this single repetitive task: turning a supplier message into a transparent purchase-order discrepancy check and a reviewable follow-up. It is designed for businesses with limited procurement software, technical support and training time. The initial evidence is a working synthetic-data prototype; SME field trials and measured savings remain future validation work.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SupplyPilot AI is a live, Orbyte-compatible workflow prototype for supplier confirmation reviews. A staff member pastes a supplier message linked to a purchase order (trigger and connected data). The Express backend calls OpenAI gpt-5-mini through Replit AI Integrations to extract product quantities, availability statements, dates and verbatim evidence from the message. The extracted facts are validated using Zod and checked against the original text. Deterministic TypeScript rules, not the language model, compare these facts with the order and flag shortages, unavailable items, late delivery or missing confirmation.

The application presents a line-by-line comparison, original-message evidence and an editable follow-up draft. A simulated Staff/Owner workflow routes a consequential decision for Owner approval or rejection. The approved message can be copied, but is never sent automatically. Review actions are shown in an Activity Log held in browser localStorage for this MVP.

If model access fails, the interface clearly labels its rule-based fallback. Process-local daily AI limits, per-IP request limits and duplicate-request protection reduce demo misuse; they are not billing guarantees. React, TypeScript, Vite, Node.js and Express support the app, which is deployed on Replit Autoscale. Workshop-relevant capabilities applied include GenAI API integration, structured validation and cloud deployment. The demo uses synthetic data and does not require live Orbyte access.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

ARCHITECTURE (deployed prototype)
Staff pastes synthetic supplier note + fixed PO-1024
-> React / TypeScript UI (Replit-hosted web artifact)
-> Express POST /api/analyze-message
-> Replit AI Integrations -> OpenAI gpt-5-mini
-> Zod schema + evidence-in-source validation
-> deterministic TypeScript PO discrepancy rules
-> editable draft -> simulated Owner approval
-> visible activity record in browser localStorage.

Stack: React + Vite frontend; Node.js + Express API; TypeScript business logic; Zod shared contracts; pnpm workspace. Hosting: Replit Autoscale (public HTTPS), one web service serving both the built UI and /api on the same port. Database: none for the current demo. Orbyte integration: proposed reusable procurement module with simulated interfaces, not a live Orbyte connection.

OFFICIAL SPONSOR RESOURCE
Replit is used for AI-assisted development, managed model access and production cloud deployment. RunPod was not used; this project runs no self-managed GPU workload.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathize / define: the Domain 2 brief identifies fragmented chat-based operations and the need for reliable, low-training automation. The project focuses on one job: checking supplier confirmations before delivery, not building a general-purpose SME chatbot. No claim is made that interviews with SME owners have already been completed.

Ideate / prototype: the workflow maps the path from message to purchase-order comparison to a human-controlled follow-up. The interface deliberately distinguishes explicit shortages from unknown details and displays excerpts supporting each AI-extracted fact. Staff can adjust the draft while the Owner makes the simulated approval decision.

Test / refine: synthetic scenarios cover full confirmation, flour shortage and delivery delay, unavailable sugar, and incomplete replies. Schema/evidence checks, a labeled rule-based fallback, request limits and local activity persistence were added. Automated frontend and API suites (29 checks at the last reported run), production builds and the publicly deployed AI workflow were tested. For Phase 2, small-business users would be observed, completion time/error rate would be measured, and real authentication plus durable audit storage would be introduced.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2 - higher economic productivity through technological upgrading and innovation. SDG 8.3 - support productive activities, entrepreneurship and the growth of micro, small and medium-sized enterprises. Source: UN SDG 8 targets (sdgs.un.org/goals/goal8); Domain 2 brief.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Planning baseline (assumption; not field-measured): 10 minutes of staff time per supplier confirmation check, including reading the note, comparing 3 PO lines and drafting a response. Measurement plan: time 30 synthetic or consented SME review tasks before/after use. Record discrepancies missed. This is a proposed baseline, not an observed trial result.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: median staff minutes per supplier confirmation check (message receipt to approved draft). Secondary KPI: % of known quantity/date issues correctly flagged on a labeled test set; % of decisions requiring human approval. Baseline to validate: 10 min/check (planning assumption).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 target: >=30% reduction in median review time (10 to <=7 min per check), compared with the validated baseline. Accuracy target: >=90% issue detection on >=30 labeled synthetic messages, with all consequential outbound messages still requiring human approval. Targets are hypotheses to be tested, not achieved impact claims.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / ReactDOM	Frontend framework	MIT
Express	Backend API framework	MIT
Vite	Web bundler / tooling	MIT
Zod	Schema validation	MIT
OpenAI gpt-5-mini	Hosted pretrained model via Replit	Provider API terms

Also used: TypeScript (Apache-2.0), Node.js (MIT), Tailwind CSS (MIT). Versions per pnpm-lock.yaml.

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24 October 2026